Benchmark overview

scCCVGBen — Single-cell CCVGAE Benchmark

scCCVGBen is a curated benchmark suite of **200 single-cell omics datasets** (100 scRNA-seq + 100 scATAC-seq) designed to evaluate the CCVGAE family of methods along three axes: **14 graph encoder variants**, **5 graph construction choices**, and **13 dimensionality-reduction baselines**.

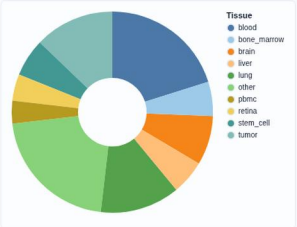
This site documents the *benchmark itself* — the datasets, the compared methods, and the evaluation metrics. For the numerical results and the statistical analysis, please refer to the accompanying paper.

Benchmark at a glance

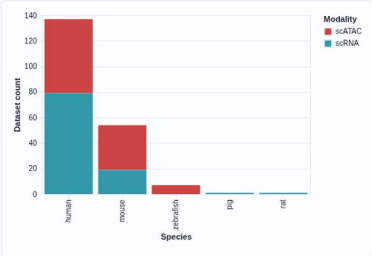
200	32	26	737,280
DATASETS	METHODS	METRICS	LARGEST CELL COUNT

Benchmark composition

Tissue distribution



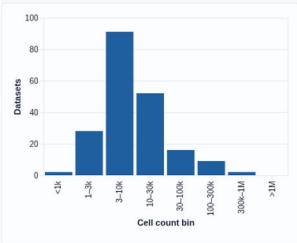
Species x modality



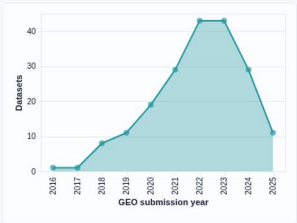
Metadata distributions (4 charts)

Metadata distributions

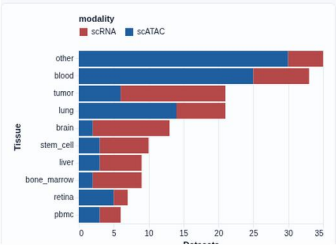
Cell-count distribution



Submission year timeline



Tissue × modality



Top 15 studies by cell count

Method catalog (32 clickable cards)

Methods

Every method compared in scCCVGBen falls into one of three categories. The full list below is auto-generated from `site/data/methods.json` (see `scripts/build_site_data.py`).

scCCVGBen graph encoders (14)

- **GAT** ATT — Graph Attention Network (Veličković 2018)
- **GATv2** ATT — Dynamic attention Graph Attention v2 (Brody 2022) — scCCVGBen extension
- **Transformer** ATT — TransformerConv — Attention is All You Need graph variant (Shi 2020)
- **SuperGAT** ATT — Self-supervised edge prediction GAT (Kim 2020) — scCCVGBen extension
- **GCN** MP — Graph Convolutional Network (Kipf 2017)
- **SAGE** MP — GraphSAGE inductive (Hamilton 2017)
- **Graph** MP — GraphConv — general message-passing variant
- **Cheb** MP — ChebNet — Chebyshev polynomial filters (Defferrard 2016)
- **TAG** MP — Topology-Adaptive Graph Conv (Du 2017)
- **ARMA** MP — ARMA filter Conv (Bianchi 2021)
- **SG** MP — Simplified Graph Conv (Wu 2019)
- **SSG** MP — Simple Spectral Graph Conv (Zhu 2021)
- **GIN** MP — Graph Isomorphism Network (Xu 2019) — scCCVGBen extension
- **EdgeConv** MP — Dynamic Edge Conv (Wang 2019) — scCCVGBen extension

Graph construction methods (5)

- **kNN_euclidean** GRAPH — Standard k-NN with Euclidean distance (k=15) — CCVGAE benchmark default
- **kNN_cosine** GRAPH — k-NN with cosine similarity — rewards direction, invariant to magnitude
- **snn** GRAPH — Shared Nearest Neighbour — 2 cells connected if they share a fraction of neighbours
- **mutual_knn** GRAPH — Mutual k-NN — only edges where both cells are in each other's k-NN list; stricter connectivity
- **gaussian_threshold** GRAPH — Gaussian heat-kernel weights $w=\exp(-d^2/(2\sigma^2))$; edges pruned at threshold 0.9

Dimensionality-reduction baselines (13)

- **PCA** BASE — Linear PCA — sklearn.decomposition.PCA
- **KPCA** BASE — Kernel PCA (RBF) — sklearn.decomposition.KernelPCA
- **ICA** BASE — Independent Component Analysis (FastICA) — sklearn.decomposition.FastICA
- **FA** BASE — Factor Analysis — sklearn.decomposition.FactorAnalysis
- **NMF** BASE — Non-negative Matrix Factorisation — sklearn.decomposition.NMF
- **TSVD** BASE — Truncated SVD — sklearn.decomposition.TruncatedSVD
- **DICL** BASE — Dictionary Learning — sklearn.decomposition.DictionaryLearning
- **scVI** BASE — Single-cell Variational Inference — Lopez 2018
- **DIP** BASE — DIP-VAE disentangled autoencoder — Kumar 2017
- **INFO** BASE — InfoVAE — Zhao 2017
- **TC** BASE — β-TCVAE — Chen 2018
- **highBeta** BASE — Hyper-parameterised VAE with high β (β=100)
- **CCVGAE** BASE — Reference CCVGAE — upstream baseline; scCCVGBen extends this with 12 encoder + 5 graph variants

Per-dataset detail page

GSE117988_MCCPBMCCancer

Field	Value
filename_key	`GSE117988_MCCPBMCCancer`
GSE / GSM accession	GSE117988
Modality	scRNA
Species	human
Tissue	pbmc
Cell count	12,874
PubMed	PMID:30250229
Submission date	Aug 01 2018
Source name (GEO)	Tumor

Tissue and species composition

Dataset index with filters

Dataset Index

Browse all 200 benchmark datasets. Use the controls below to filter by modality, category, or tissue; the table is also sortable by every column.

Each row links to a per-dataset detail page with GEO-verified metadata (species, tissue, submission date, PubMed, source name).

Modality: All Category: All Tissue: All

Show 25 entries Search:

Key	GSE	Modality	Tissue	Species	Category	Cells	Description
GSE128033_new	GSE128033	scRNA	immune	human	immune	737,280	GEO cellranger (737280 cells)
GSE247719_PanSci_05_Muscle_adata	GSE247719	scRNA	muscle	mouse	muscle	696,410	GSE247719_PanSci_05_Muscle_adata
GSE164378_new	GSE164378	scRNA	pbmc	human	pbmc	161,764	GEO cellranger download (161764 cells)
GSE262288_breastMetasisHmCancer	GSE262288	scRNA	tumor	human	tumor	157,518	GSE262288_breastMetasisHmCancer
GSE174367_snATAC-seq_filtered_peak_bc_matrix	GSE174367	scATAC	brain	human	brain	143,401	GSE174367_snATAC-seq_filtered_peak_bc_matrix
GSE222369_NKsLymphomaHmCancer	GSE222369	scRNA	tumor	human	tumor	138,805	GSE222369_NKsLymphomaHmCancer
GSE235787_bcellsALLHmCancer	GSE235787	scRNA	tumor	human	tumor	113,495	GSE235787_bcellsALLHmCancer
GSE181061_new	GSE181061	scRNA	kidney	human	kidney	108,421	GEO cellranger download (108421 cells)
GSE190604_new	GSE190604	scRNA	ovary	human	ovary	103,805	GEO cellranger (103805 cells)
GSE189070_astrocytesSCIMmDev	GSE189070	scRNA	brain	mouse	brain	103,446	GSE189070_astrocytesSCIMmDev
GSE120926_new	GSE120926	scRNA	tumor	human	tumor	102,525	GEO cellranger (102525 cells)
GSE253355_bmNicheHm	GSE253355	scRNA	bone_marrow	human	bone_marrow	99,846	GSE253355_bmNicheHm
bm_GSE120446	GSE120446	scRNA	bone_marrow	human	bone_marrow	90,653	bm_GSE120446
GSE149689_new	GSE149689	scRNA	blood	human	blood	85,144	GEO cellranger download (85144 cells)
GSE225600_breast_CancerHm	GSE225600	scRNA	tumor	human	tumor	81,683	GSE225600_breast_CancerHm
GSE222002_TcellsHmCancer	GSE222002	scRNA	tumor	mouse	tumor	78,315	GSE222002_TcellsHmCancer
GSE183904_GastricHmCancer	GSE183904	scRNA	tumor	human	tumor	62,035	GSE183904_GastricHmCancer
GSE145929_ProgastinMmDev	GSE145929	scRNA	bladder	mouse	bladder	61,014	GSE145929_ProgastinMmDev
GSE151530_new	GSE151530	scRNA	liver	human	liver	56,721	GEO cellranger download (56721 cells)
GSE123813_bccHmCancer	GSE123813	scRNA	tumor	human	tumor	53,030	GSE123813_bccHmCancer
GSE148218_bmALLHmCancer	GSE148218	scRNA	tumor	human	tumor	50,747	GSE148218_bmALLHmCancer
GSE189357_lungAdreHmCancer	GSE189357	scRNA	lung	human	lung	45,938	GSE189357_lungAdreHmCancer
GSE123902_LungAdreHmCancer	GSE123902	scRNA	lung	human	lung	42,847	GSE123902_LungAdreHmCancer
GSE132509_acutelymluekPBMCHmCancer	GSE132509	scRNA	pbmc	human	pbmc	39,375	GSE132509_acutelymluekPBMCHmCancer
GSE138709_LiverCancer	GSE138709	scRNA	liver	human	liver	33,991	GSE138709_LiverCancer

Showing 1 to 25 of 200 entries Previous 1 2 3 4 5 … 8 Next

Table data comes from `data/benchmark_manifest.csv`, regenerated by `scripts/build_site_data.py` (tissue canonicalised via keyword scan over GEO source_name + description + title).

Metric registry (26 with details)

Evaluation Metrics

scCCVGBen scores every (dataset, method) pair on 26 metrics grouped into three families. The list below is auto-generated from `site/data/metrics.json`.

Clustering quality (6)
External labels, cluster separation, and original-space distance preservation.
`ASW` `DAV` `CAL` `NMI` `ARI` `COR`

Dimensionality-reduction evaluation — UMAP + tSNE (10)
Neighbourhood preservation and global structure after UMAP/tSNE projection.
`distance_correlation_umap` `Q_local_umap` `Q_global_umap` `K_max_umap` `overall_quality_umap` `distance_correlation_tsne` `Q_local_tsne` `Q_global_tsne` `K_max_tsne` `overall_quality_tsne`

Latent-space intrinsic geometry (10)
Intrinsic latent geometry, robustness, trajectory signal, and interpretation.
`manifold_dimensionality_intrin` `spectral_decay_rate_intrin` `participation_ratio_intrin` `anisotropy_score_intrin` `trajectory_directionality_intrin` `noise_resilience_intrin` `core_quality_intrin` `overall_quality_intrin` `data_type_intrin` `interpretation_intrin`

Clustering quality (6)

Metric	Description	Source	Direction
ASW	Average Silhouette Width	sklearn.metrics.silhouette_score	Higher is better
DAV	Davies–Bouldin Index	sklearn.metrics.davies_bouldin_score	Lower is better

Per-method detail page

GATv2

Field	Value
Family	attention
Group	scCCVGBen graph encoder

Description

Dynamic attention Graph Attention v2 (Brody 2022) — scCCVGBen extension

Role in scCCVGBen

Axis A (encoder-variation) sweep: CCVGAE trains a latent representation with this message-passing / attention module while holding the graph fixed to k-NN Euclidean. Benchmark naming for sweep rows: `scCCVGBen_GATv2`.

Auto-generated from `scripts/build_site_data.py`. scCCVGBen benchmark tests each method against the same 200 datasets and 26 metrics; see the Methods index for the full set.